Ready-to-use paragraphs for paper

Results section:

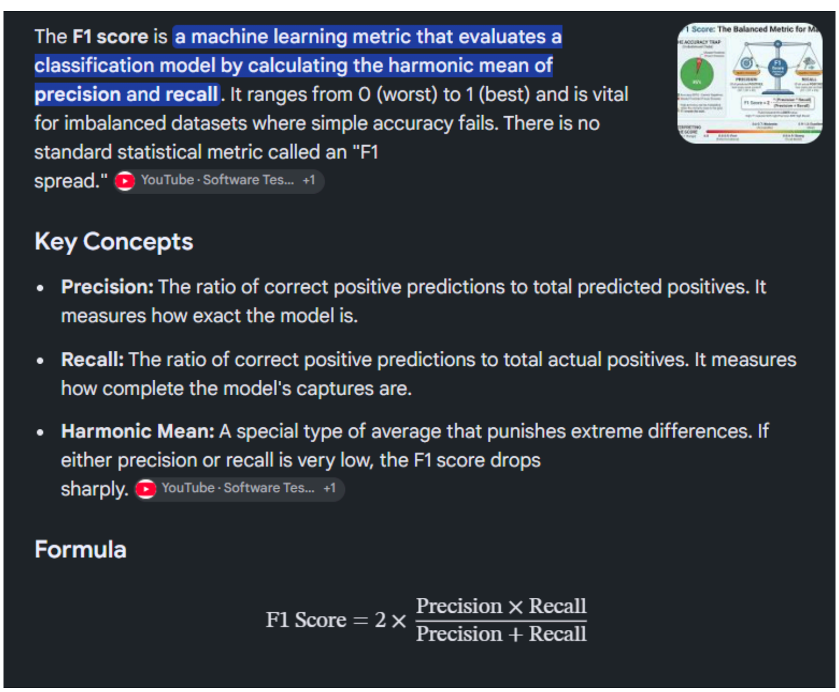
Table 1 summarizes the performance of the four machine learning algorithms evaluated. Among the models tested, K-Nearest Neighbors (KNN) achieved the highest performance across all four metrics, with an accuracy of 90.2%, precision of 93.3%, recall of 87.5%, and F1-score of 90.3%. Support Vector Machine (SVM) ranked second (86.9% accuracy), followed by Logistic Regression (85.2%) and Random Forest (83.6%).

The confusion matrix for the best-performing model, KNN (Figure 2), shows that out of 61 test patients, the model correctly identified 27 of 29 patients without heart disease and 28 of 32 patients with heart disease, with 2 false positives and 4 false negatives.

Discussion section:

Given the clinical nature of this task, false negatives are of particular concern, as they represent at-risk patients who could be overlooked. The KNN model's recall of 87.5% indicates reasonably strong, though not perfect, sensitivity to positive cases.

Feature importance analysis using Random Forest (Figure 3) identified oldpeak, maximum heart rate achieved, number of major vessels, and chest pain type as the most influential predictors. These findings align with established clinical understanding, as exercise-induced ST depression and abnormal heart rate response are well-documented cardiovascular risk indicators. Notably, KNN's strong performance suggests that patients with heart disease in this dataset exhibit closely clustered patterns across measured variables, making similarity-based reasoning particularly effective here. Given the relatively small test set (61 patients), these results should be treated as indicative rather than conclusive.



The **F1 score** is a machine learning metric that evaluates a classification model by calculating the harmonic mean of precision and recall. It ranges from 0 (worst) to 1 (best) and is vital for imbalanced datasets where simple accuracy fails. There is no standard statistical metric called an "F1 spread." YouTube · Software Tes... +1

Key Concepts

- **Precision:** The ratio of correct positive predictions to total predicted positives. It measures how exact the model is.
- **Recall:** The ratio of correct positive predictions to total actual positives. It measures how complete the model's captures are.
- **Harmonic Mean:** A special type of average that punishes extreme differences. If either precision or recall is very low, the F1 score drops sharply. YouTube · Software Tes... +1

Formula

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

I worked out the numbers by hand from your table too, so here's what you should see, and what it means:

Model	Smallest spread across hospitals	What's happening
SVM	**0.050** (most stable)	F1 stays in a tight 0.749–0.799 band everywhere
Random Forest	0.061	Also very steady, 0.740–0.801
KNN	0.085	Reasonably steady, 0.693–0.778
Logistic Regression	**0.227** (least stable, by far)	Falls apart specifically on Switzerland

This is your actual finding, and it's a good one. Look closely at Logistic Regression on Switzerland: precision stayed high (0.978 — when it says "disease," it's almost always right) but recall crashed to 0.391 — meaning it missed the majority of genuinely sick patients there. SVM and Random Forest, tested on that exact same Switzerland data, kept recall around 0.65 — far from perfect, but nowhere near as broken.

In plain words for your Discussion section: the straight-line, formula-based method (Logistic Regression) is the one that breaks down most when patient populations differ. The boundary-drawing method (SVM) and the vote-based method (Random Forest) hold up noticeably better. That's a real, specific, defensible answer to "which model is most reliable across hospitals" — not a vague accuracy comparison, but a concrete mechanism you can explain.

One honest note to include: Switzerland's extreme imbalance (115 vs. 8, from a few steps ago) is likely part of *why* it's the hardest hospital for every model — worth saying so plainly rather than let it look mysterious.

Run the code above to confirm the exact printed numbers, paste it back, and then we'll move to Step 8: a chart showing this spread visually — the clearest way to make this finding land in the paper.